

Master Theory A-Z

Classification, Decision Trees, KNN, Lazy Learning, Pruning, KNN Imputation

Learning Day 7B - Classification Missing Coverage Bundle

1. Executive Summary

This master theory document consolidates the missing Learning Day 7B classification coverage. It combines foundations, Decision Tree theory and pruning, KNN theory, KNN practical modelling, and KNN imputation into one A-to-Z learning reference.

The goal is not only to list definitions. The goal is to explain how each concept connects to the customer segmentation project and how the concepts were applied in the notebooks.

2. Final Coverage Map

Notebook / Source Area	Coverage Status	Main Content
Notebook 1 - Classification Foundations	Completed	Supervised learning, classification vs regression, features and target, binary/multiclass/multilabel, algorithm families, compact EDA.
Notebook 2 - Decision Tree Theory and Pruning	Completed	Tree structure, Gini, Entropy, Information Gain, Gain Ratio, bias, pruning, decision boundaries, overfitting.
Notebook 3 - KNN Practical and Lazy Learning	Completed	Lazy vs eager learning, KNN theory, distance, scaling, KNN model, k tuning, voting, limitations, KNN imputation.
Original Decision Tree MVD Notebook	Completed earlier	DecisionTreeClassifier, max_depth, Gini vs Entropy, min_samples_leaf, ccp_alpha pruning, feature importance, SHAP/oblique attempt.

3. Classification Foundations

Classification is supervised learning where the target is a category. In this project the target is Segmentation, with four possible classes: A, B, C, and D.

Concept	Simple Meaning	Project Example
Feature	Input information used by the model.	Age, Profession, Spending_Score, Family_Size.
Target	Correct answer to predict.	Segmentation.
Class	One possible target category.	A, B, C, D.
Prediction	Model guessed class.	Predicted customer segment.
Train/test split	Separate learning data from evaluation data.	80% train, 20% test with stratify.

4. Binary, Multiclass, and Multilabel Classification

Type	Simple Meaning	Example
Binary classification	Only two possible labels.	Fraud / Not Fraud.
Multiclass classification	More than two labels, but only one correct label per row.	Customer Segment A/B/C/D.
Multilabel classification	One row can have more than one correct label.	Customer interests: Sports + Travel.

The customer segmentation problem is multiclass classification because each customer belongs to one of A, B, C, or D.

5. Dataset and EDA Summary

The same Kaggle customer segmentation dataset was used across the missing notebooks. It contains 8,068 rows and 11 columns in Train.csv. The target column is Segmentation.

Finding	Value / Pattern	Meaning
Target balance	A: 24.44%, B: 23.03%, C: 24.42%, D: 28.11%	Reasonably balanced target.
Average age	D: 33.4, A: 44.9, B: 48.2, C: 49.1	Segment D is younger on average.
Spending score	D: 87.04% Low, C: 45.84% Average	Spending behavior helps segment separation.
Var_1	Cat_6 appears in all segments, highest in C at 76.76%	Useful but anonymous business meaning.

6. Decision Tree Theory

A Decision Tree predicts by asking questions step by step. It starts at the root node, follows branches through internal nodes, and ends at a leaf node that gives the prediction.

Tree Part	Simple Meaning
Root node	First question asked by the tree.
Internal node	A later question after the root.
Branch	The path followed after answering a question.
Leaf node	Final prediction point.

Decision Trees are useful for business interpretation because their logic can be written as if-then rules.

7. Split Quality: Gini, Entropy, Information Gain, Gain Ratio

Concept	Simple Meaning	Best Direction
Gini Index	Measures how mixed a group is.	Lower is better.
Entropy	Measures confusion inside a group.	Lower is better.
Information Gain	Measures how much confusion a split removes.	Higher is better.
Information Gain Bias	Information Gain may prefer columns with too many unique values.	Avoid ID-like columns.
Gain Ratio	Corrects the bias toward too many tiny groups.	Useful split, not just clean split.

A good split creates cleaner child groups and remains useful for future data. A bad split may memorize the training rows, such as customer IDs.

8. Greedy Search, Hypothesis Space, and Inductive Bias

The hypothesis space is the set of all possible trees the model could build. Decision Trees cannot try every possible full tree, so they use greedy search: choose the best-looking split now and continue forward.

Term	Simple Meaning
Hypothesis space	All possible trees the model could build.

Term	Simple Meaning
Greedy search	Choose the best split at the current step.
No backtracking	Usually do not go back and change earlier splits.
Inductive bias	The model preference for cleaner splits and simpler rules.
Occam Razor	If two trees perform similarly, prefer the simpler one.

9. Decision Boundaries and Decision Regions

A Decision Tree divides the feature space into decision regions. With numeric features, standard tree splits are usually vertical or horizontal cuts. This makes tree regions box-like and rule-based.

Business meaning: a tree may divide customers by age, spending score, family size, or work experience and assign different segments to different regions.

10. Overfitting and Pruning

Overfitting happens when a tree learns training data too closely and performs worse on unseen data. Pruning controls tree size to improve generalization.

Control / Concept	Simple Meaning	Why It Helps
max_depth	Maximum number of levels.	Stops tree from becoming too deep.
min_samples_split	Minimum rows needed before a node can split.	Stops tiny groups being split too much.
min_samples_leaf	Minimum rows in a final leaf.	Prevents unstable tiny prediction groups.
ccp_alpha	Cost-complexity pruning strength.	Removes branches that add complexity without enough benefit.
Subtree replacement	Replace branch with one leaf.	Simplifies tree.
Subtree raising	Move useful lower branch higher.	Simplifies structure.
Error-based pruning	Remove branch if errors do not increase much.	Keeps useful branches only.
MDL idea	Prefer shorter explanation if performance is similar.	Keeps model understandable.

11. KNN Theory

KNN stands for K-Nearest Neighbors. It predicts a new customer by looking at the most similar old customers and using their known segments to vote.

Concept	Simple Meaning
k	Number of nearest neighbors used for prediction.
Neighbor	An old customer similar to the new customer.
Distance	How close or far two customers are.
Euclidean distance	Straight-line distance between numeric feature values.
Majority voting	Every neighbor vote counts equally.
Distance voting	Closer neighbors receive stronger vote.

12. Lazy vs Eager Learning

Model	Learning Style	Training Behavior	Prediction Behavior
Decision Tree	Eager learner	Builds rules during training.	Follows rules quickly.
KNN	Lazy learner	Stores training examples.	Compares new customer with old customers.

The timing demo confirmed the concept: KNN trained quickly but predicted slower, while Decision Tree trained longer but predicted quickly.

13. KNN Preprocessing

KNN needs clean numeric input because it uses distance. Missing values, text categories, and different numeric scales must be handled before training.

Column Type	Preprocessing	Reason
Numeric	Median imputation + StandardScaler	Fill missing values and make scale fair.
Categorical	Most frequent imputation + OneHotEncoder	Convert text categories into numeric 0/1 columns.
Target	Keep as segment labels A/B/C/D	This is the predicted answer.

14. KNN Model Results

Experiment	Training Accuracy	Test Accuracy	Train-Test Gap	Interpretation
Baseline KNN k=5	0.6264	0.4796	0.1469	Some overfitting.
Best KNN k=21 uniform	0.5555	0.5204	0.0350	Best generalization in this experiment.
k=21 distance voting	0.9558	0.5025	0.4534	Strong overfitting.

Best KNN configuration: n_neighbors = 21 and weights = uniform.

15. KNN Imputation

KNNImputer fills missing numeric values by looking at similar rows. It can create more customized estimates than median imputation, but it is slower and depends on distance quality.

Method	Simple Behavior	Strength	Risk
Median imputation	Fill with one middle value.	Simple and stable.	Less customized.
KNN imputation	Estimate using similar rows.	More customized.	Needs numeric scaled data and can be slower.

16. KNN vs Decision Tree

Area	Decision Tree	KNN
Core logic	Learns if-then rules.	Finds similar customers.
Learning style	Eager learner.	Lazy learner.
Scaling needed	No.	Yes.

Area	Decision Tree	KNN
Interpretability	High; clear rules.	Medium; similar-case explanation.
Main weakness	Can overfit if too deep.	Sensitive to scale, noise, irrelevant features, many dimensions.

17. Final Learning Conclusion

The missing Learning Day 7B coverage is complete. The final structure now includes theory, practical modelling, business interpretation, model tuning, and clear comparison between rule-based and distance-based classification.

The best learning takeaway is: Decision Trees explain with rules; KNN explains with similar examples. Both can be useful, but they require different preprocessing and interpretation approaches.